Human African Trypanosomiasis in the Democratic Republic of the Congo

Introduction:
West African HAT, caused by the parasite *Trypanosoma brucei gambiense,* is transmitted by tsetse flies in a human-fly-human cycle. Unlike East African HAT, caused by *Trypanosoma brucei rhodesiense*, which has cattle and some wild animals as a reservoir, West African HAT is essentially a disease of humans. It is assumed that in the absence of appropriate treatment, HAT infection inevitably leads to death.[1] West African HAT has two stages, the hemolymphatic stage with no or few specific symptoms, followed months to years later by the meningoencephalitic stage when the causative parasite has crossed the blood/brain barrier. The second stage is characterized by neurological signs and personality changes. Damage to the hypothalamus may lead to disturbance of the normal sleep pattern which has led to the disease being called ‘sleeping sickness’. [2]

About 60% of all cases of West African HAT reported worldwide are from the Democratic Republic of the Congo (DRC), which has seen a surge in cases during the second half of the 1990s. The highest peak was observed in 1998, when 26,318 cases were reported; by 2018 the reported annual incidence was down to 661 cases. (Un-published data, Programme Nationale de Lutte contre la Trypanosomiase Humaine Africaine) According to estimations for the period 2014-2018, about five million people are estimated to live in areas of moderate to high risk in DRC.[3]

Diagnostic procedures
In diagnosis of HAT there are three steps, screening, confirmation tests and staging.[4] Screening is mostly done on the basis of a serological test called the card agglutination test for trypanosomiasis (CATT), based on the LiTat1.3 antigen produced at ITM.[5] The CATT test is done on capillary blood and can easily be performed under field conditions but does require a battery operated rotator. For long term storage a cold chain is also required. It has a specificity of around 95% and a sensitivity ranging from 87-98%. More recently two rapid diagnostic tests (RDTs) have been developed, both based on native antigens produced at ITM (LiTat1.3 and LiTat1.5).[6, 7] They have the advantage of being individual, whereas the CATT is produced in vials of 50 doses that need to be used the same day. Another advantage of RDTs is that they are thermostable and do not require any equipment or source of electricity. RDTs based on recombinant antigen are still under development.

Individuals testing positive on screening tests are subjected to a number of parasitological confirmation tests. The routine confirmation tests used are microscopic examination of a lymph node aspirate (LNA), as well as the capillary tube centrifugation test (CTC)[8] and the mini-anion-exchange centrifugation technique (mAECT), both on venous on blood samples.[9] The LNA test is highly specific but has poor sensitivity, 44.8 % (95% CI 36.8–53.0) at most. To overcome the problem of low sensitivity, a number of concentration techniques have been developed. These include the CTC and the mAECT. Both methods are based on concentration of trypanosomes through centrifugation; mAECT also uses anion exchange chromatography to separate the trypanosomes, which are less negatively charged than blood cells, from venous blood. In all three techniques the final product is examined by microscopy. In LNA a wet preparation from a lymph node aspirate is examined at a magnification of 40x to identify the presence of motile Trypanosomes. In CTC and mAECT, tubes are examined directly at low magnification (10-20x). These techniques are more sensitive then LNA but still far from perfect, reported sensitivity estimates range from 56.5% (95% CI 48.3–64.5%) for CTC to 75.3% (95% CI 67.7–81.9%) for mAECT. Sensitivity can be increased by using buffy coat but this requires highly skilled laboratory staff. If the wrong fraction is sampled and examined, sensitivity will not increase but more likely drop to almost zero. CTC is cheap but requires a hematocrit centrifuge. Contrary to ordinary battery operated centrifuges, battery operated hematocrit centrifuges are hard to find. Another disadvantage of the CTC is that one examines the full depth of a capillary tube, it’s hard to focus and there is a risk of confusing artifacts with parasites.[10] All three confirmation tests are done on unstained live parasites, visible because they move. Slides/tubes cannot be kept for quality assurance because the parasite will soon die and disintegrate.

Once the trypanosome has been demonstrated, staging of the disease is the next step. This requires a lumbar puncture. Cerebrospinal fluid is examined for the presence of trypanosomes or a raised white blood cell count. If either of the two is present, the patient is assumed to be in the meningo-encephalitic stage. Staging is necessary because the treatment for the meningo-encephalitic stage is different from the treatment of the hemolymphatic stage and until recently was highly toxic.

Treatment

In the hemolymphatic stage, West African HAT is treated with daily intramuscular injections of pentamidine for 1 week. In the meningo-encephalitic stage until fairly recently (2010) treatment has been with melarsoprol, a trivalent arsenic compound. It was given by slow intravenous injection either over a period of 10 consecutive days, or in 3-4 series spread out of over one month. The most serious side effect associated with melarsoprol is encephalopathy, which usually manifests itself during treatment as a sudden violent neurological deterioration. Lethal forms of encephalopathy can be expected in 3-5% of cases treated.[11] Apart from these obvious toxicity problems, treatment with melarsoprol has also been shown to have high failure and relapse rates.[12]

More recently some less toxic treatments have become available. Di-fluoro-methyl-ornithine (DFMO) was first used for trypanosomiasis in 1985.[13] The drug is administered by intravenous infusion four times a day for two weeks. During a trial that started in 2003, a combination of DFMO and nifurtimox (NECT) has been shown to be very effective and far less toxic than melarsoprol. The NECT schedule consists of twice daily administration of DFMO by intravenous infusion for 7 days, combined with daily oral administration of nifurtimox for 10 days.[14] Since 2010 NECT has been adopted as the first line treatment by most HAT control programs. It is a major improvement on Melarsoprol but still not a field friendly treatment since it requires twice daily IV infusions for 7 days, difficult to implement at health center level.

In 2020 an oral drug, Fexinidazole (once daily intake, during 10 days), has become available that is effective in both stages of the disease. It is recommended by WHO for individuals ≥6years with a body weight of ≥20 kg, who have first or second stage, but with <100 leucocytes per µl CSF. For clinically serious cases NECT is still the preferred treatment because of a slightly higher effectiveness. In practice this means that lumbar puncture is no longer recommended for patients that present no signs of severe second stage disease [15]. The real breakthrough may come soon once Acoziborole becomes available, currently the phase 3 evaluation is in its final stages.[16] Acoziborole is a single dose oral treatment, apparently of low toxicity, and could therefore be considered even without parasitological confirmation in case of a positive CATT or RDT. However there will always remain a need to monitor which proportion of persons treated presumptively truly are HAT cases. Currently the positive predictive value of the screening tests is in the order of 1-2% and this is likely to decrease even further as incidence goes further down.

Control strategy

The backbone of HAT control has always been active screening of the population at risk combined with free treatment for all cases identified. The aim is to achieve a reduction of transmission by eliminating the parasite from the human reservoir. Historically a number of West African HAT foci have been brought under control this way.[17-19] Active case finding for HAT in the DRC is traditionally based on mobile teams visiting endemic villages and screening the entire population using the CATT test. Those testing positive are subjected to confirmation tests on the spot. Cases identified undergo a lumbar puncture for disease staging; they are then referred to fixed facilities for treatment. Difficulties in securing sufficient participation in the population screening sessions, and the deficient sensitivity of screening algorithms, especially when it comes to the confirmation tests used, appear to be important bottle necks.[20, 21]

Vector control is another option, it can be based on area wide measures which include aerial spraying and sterile insect technique, or more localized measures such as the use of traps.[22, 23] Though vector control yielded good results in research settings, for a long time doubts have remained about its feasibility and sustainability; it is expensive and requires active cooperation from the local population. In 2001 the Pan African Tsetse and Trypanosomosis Eradication Campaign (PATTEC) was launched by the Organization of African Unity. The campaign aims at eradicating the disease as well as the tsetse fly vector from the African continent.[24] A more feasible approach to vector control has been developed by the Liverpool School of Tropical Medicine (LSTM) and the Institut de Recherche et de Développement (IRD), who started using 'tiny targets'. Tiny targets are insecticide treated screens that measure 25 x 50 cm and can be mass produced for less than 1 USD a piece. [25] They are particularly effective against riverine tsetse species and have been shown to reduce vector density by as much as 90%.

The success of control efforts since the late 1990s made the World Health Organization (WHO) decide to target HAT elimination as a public health problem by 2020 and zero incidence of the disease by 2030. [26] In 2018, 977 new HAT cases were reported, with over 75% of the patients identified in the Democratic Republic of Congo (DRC).

HAT in DRC

The history of HAT in the DRC has been a vicious cycle of increasing prevalence, control measures being implemented, decreasing prevalence, control measures being neglected, and again increasing prevalence. At the end of the colonial era,, in 1960, the disease was under control but gradually reemerged when control measures were abandoned. During the early 1990s control measures were completely and abruptly interrupted, resulting in a major resurgence going unnoticed for a long period; by the end of the 20th century the reported incidence levels were almost back to those of the late 1920s and early 1930s (figure 1).

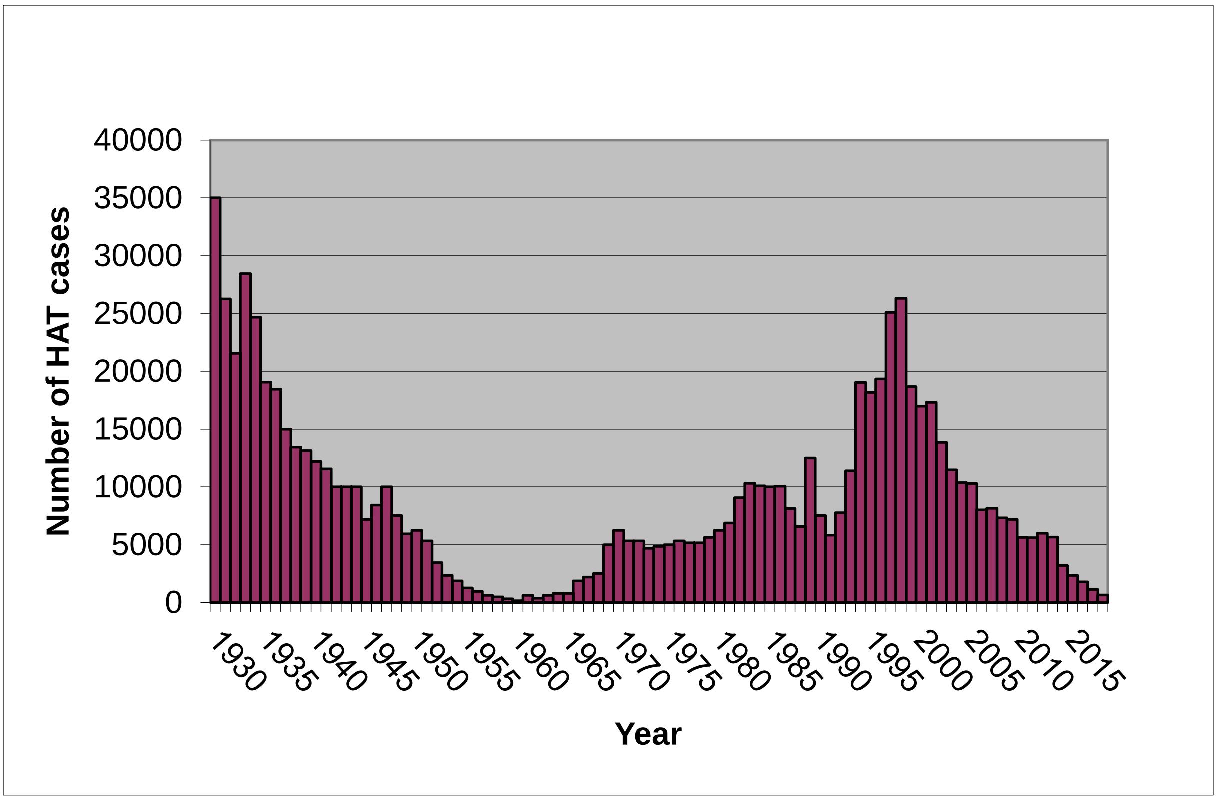

Figure 1 Evolution in reported incidence of HAT in the DRC from 1926-2019

When after 1998 control measures were resumed and intensified, the disease was rapidly brought under control again in most provinces affected. Two provinces in DRC were lagging behind, Kasai and Bandundu .[27] By 2015, Bandundu province (more recently split into Kwilu, Kwango and Mai Ndombe) accounted for 50-60% of all reported HAT cases across the continent. There have also been reports of HAT cases in areas not under surveillance. [28]

Once HAT prevalence is somewhat reduced, effectiveness and cost-effectiveness of active case finding becomes problematic. Participation rates in active screening drop because people no longer consider HAT a threat, at the same time the costs per case detected rise dramatically.[20, 29] Taking into account these constraints, the national HAT control programme, also known as *Programme National de Lutte contre la Trypanosomiase Humaine Africaine* (PNLTHA) formulated a strategy in which integration of HAT control in the district health system is one of the key components.[30] However it has been observed that patients presenting themselves to the health services at their own initiative are often misdiagnosed. This resulted in an average health systems delay of more than one year among 14 patients interviewed in Kinshasa in 2007 (Ehomba 2007, unpublished data).

The current health care system in the DRC was designed, based on the Alma Ata declaration. The system aims at providing global, integrated and continuous care, with participation of the community.[31] In the DRC the health district is called '*zone* *de santé*'; altogether there are 515 zones, with an average population of just above 120,000. In each zone there is a 'general referral hospital', run by the zonal health coordinator, who is a medical doctor. Each zonal health coordinator supervises a network of 5 or more health centres, run by nurses. Utilization rates of health care services are typically low with an estimated 0.15 consultations per inhabitant per year. In addition to the public system, a myriad of qualified and mostly unqualified private practitioners exists.

Until recently there used to be no less than 53 vertical disease control programs in the DRC but their numbers have now been reduced. One of the remaining programmes is the HAT control programme, the PNLTHA. The PNLTHA has its own staff and infrastructure; only recently the first steps towards integration of HAT case finding and treatment have been initiated. Case finding is traditionally active, involving mobile screening teams. A mobile screening team is made up of 6-8 persons, moving around by 4WD vehicle. When they arrive in a village the population are invited to be registered. The next day screening activities start. This is a time consuming process and increasingly there are indications that those most at risk, young adults, are least likely to participate. Figure 2 below shows the population pyramid of participants in screening by a mobile team in Yasa Bonga, Bandundu. Obviously school aged children are screened in large numbers but young adult men in particular are underrepresented.

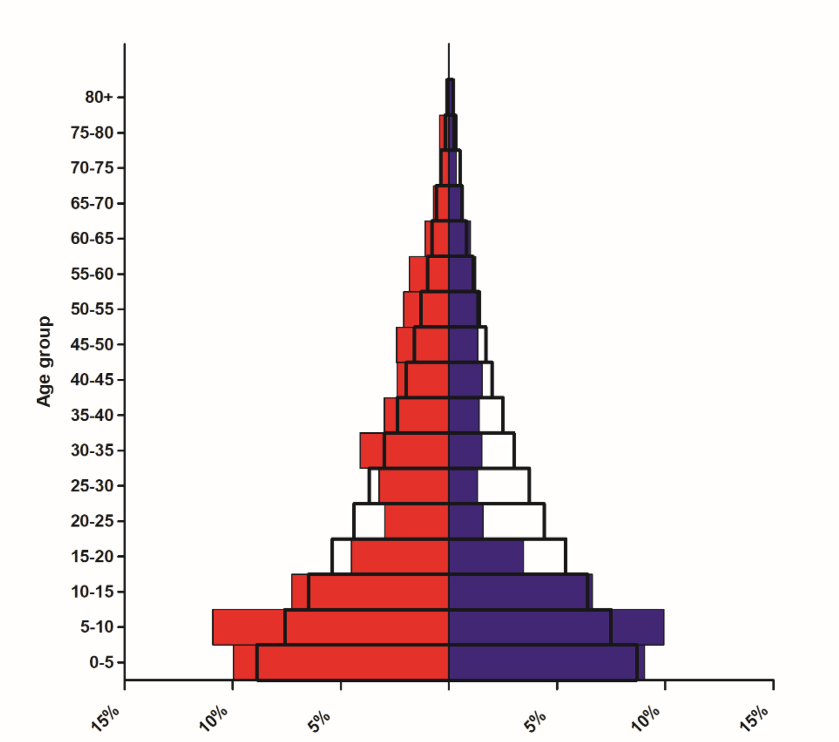

Figure 2: Participation of women (red bars) and men (blue bars) in HAT screening. The transparent bars show the population distribution.

HAT cases identified by mobile screening teams are referred for treatment to fixed facilities. Whereas previously these fixed facilities were also specialized HAT facilities, they are now gradually being integrated into general health centers.

Increasingly HAT diagnostic and treatment services are also provided from other general health centers and hospitals. However the diagnostic procedures remain complex (making it difficult to maintain expertise at decentralized level in case of very low numbers of cases) and attendance of health facilities other than hospitals is often minimal. A study by Mitashi et al in 2015, who visited all health centers in Kwamouth and Mushie (Bandundu), at the time the most HAT endemic districts of DRC, revealed that staff were well aware of HAT but that diagnostic means (microscopes, mAECT) were lacking and that health centers were poorly attended. [32] The median number of consultations in the week before the visit was 14, i.e. 2 per day. This includes all consultations, for whichever ailment.

Since 1998 the bulk of the funding for active case finding in DRC has been provided by Belgian development aid (DGD) through its technical implementation agency, the Belgian Technical Cooperation (BTC, now renamed Enabel). In 2012 the Belgian Government decided to stop its governmental cooperation to the health sector in DRC. This raised concern that if active screening was to be abandoned, the history of HAT in DRC might once again repeat itself. [33] After lobbying, DGD decided to continue funding but channelled through ITM (FA3[1], 5.5 million Euro). In line with his philosophy of 'switching the poles', i.e. making the countries in the South responsible for implementation of research and support, the then director of ITM decided to direct the funds directly to the PNLTHA, without intermediary. The PNLTHA was to develop annual action plans and budgets, ITM provided support in bookkeeping and reporting. From the start the PNLTHA were asked to rationalize activities and prepare for a future with less funding available and with a different epidemiological situation. When the contract was renewed in 2016 (FA4, 14.2 million Euro), it was explicitly stated that active screening in its current format (i.e. by the traditional large mobile teams) had to be phased out.

As active screening by mobile teams had been the mainstay of PNLTHA activities for decades, such a change ran counter to many vested interests. There were 34 mobile teams, all staff almost fully dependant on incentives paid through Belgian aid. As the number of vertical disease control programs was being reduced, many now redundant staff were sent to the PNLTHA headquarters in Kinshasa, further complicating matters. In recent years numbers of staff in the central PNLTHA office increased from approximately 50 to more than 70. Whichever decision the PNLTHA management takes is bound to be scrutinized by many different persons who usually have other priorities than the control of HAT.

The BMGF funded ITM projects.
In 2013, ITM submitted a first proposal to the BMGF (OPP1084252_ITM, budget 888,000 USD, locally named 'HATCase'). The objective was to develop and validate an alternative system of active case finding, making use of motorcycle based mini-teams passing door-to-door and screening for HAT with one of the new rapid diagnostic tests (RDTs). For any RDT positive subject a venous blood sample was collected, which was used to prepare three filter paper samples, two whole blood and one buffy coat. Whenever the teams returned to their coordination in Bandundu town, these filter papers were handed over to a lab technician who performed LAMP, an automated molecular test, on them.[34] Earlier in 2013 a phase 2 evaluation of LAMP

[1] FA3 and FA4 refers to framework agreements between ITM and DGD

on stored serum samples at ITM had shown promising results. [35] The hypothesis was that the combination of a positive serological test plus a positive molecular test would be highly specific and have a high positive predictive value. LAMP was performed on one of the whole blood samples and on the buffy coat samples, the second whole blood sample was sent to ITM for quality assurance by Immune Trypanolysis (ITL). ITL is a test that is highly sensitive and extremely specific and is used as gold standard for antibody detection.[36] Unfortunately the test is too complex and too cumbersome to be applied at peripheral level. Subjects positive on RDT and LAMP were to be referred to a nearby fixed health facility for diagnostic confirmation. In addition we introduced digital data capturing tools and recorded GPS coordinates of all persons screened.

The strategy worked reasonably well and when Matt Steele visited the project sites in July, 2014, he was very positive. But some problems had become apparent. There appeared to be an issue with the LAMP, HAT cases were found among LAMP negatives who presented for diagnostic confirmation because of symptoms. Also it became clear that preparing buffy coat in the field was too cumbersome and might be the reason for false negatives if the fraction of the sample taken was not correct. In addition the RDTs were less specific than hoped for (±95%), leading to a high workload both in the field for preparing filter paper samples and in the lab where the LAMP was done. Furthermore we noticed that although the specificity of the combination of RDT + LAMP was very high, 99.6%, people living at more than 5 kilometer from the designated health center would not come for confirmation testing. At the same time it was observed that specificity of the CATT test used by the traditional mobile teams in the area was almost as high as that of the combination of RDT + LAMP, i.e. 99.5%. Out of over 550,000 persons screened by seven mobile teams in the preceding year, only 0.5% had been found CATT positive and among those 64% had been confirmed as HAT cases. Countrywide the positive predictive value of CATT used in mobile teams was 23%. This prompted us to reconsider our approach and two alternatives were subsequently tried out.

In Beno health zone we used CATT and RDT in parallel as screening tests and filter papers were collected for anyone testing positive on either of the two tests. Upon completion of initial screening activities and after LAMP results had also become available a lab technician on motorcycle was sent with a microscope to perform diagnostic confirmation for each suspect identified, irrespective of the LAMP result. He was also asked to collect an additional filter paper sample (whole blood only) for each suspect tested for Quality Assurance (QA) by ITL at ITM. In addition we had developed an adaptor that could be fitted on a trinocular microscope and that could be used to support the smartphone used for digital data collection (figure 3). Thus the technician was able to record a short video of what he saw through the microscope in case of a positive confirmation test. We had become a bit suspicious of the extremely high positive predictive value of CATT reported by the traditional mobile teams. But the slides or collector tubes cannot be kept for QA because the parasites will soon be dead and disintegrate. A video with time stamp, an option included in the app used, was to provide strong evidence that actually parasites had been seen.

Figure 3: Trinocular microscope with adaptor to mount smartphone for recording video

In Mbaye Larème health zone we tested another strategy, three individual screeners on motorcycles performing door-to-door screening using CATT, followed up after a short delay by a microscopist also on motorcycle to perform diagnostic confirmation on the spot.

In the meantime Matt Steele had been contacted by the Mary A. Cargill Foundation (MACF) who were looking for a healthcare project to support. This resulted in an additional grant of 5 million USD (later named ‘Kwilu project’), which we decided to use to reinforce the health system in two HAT endemic zones in Bandundu (Yasa Bonga and Mosango) and to introduce active screening by mini-teams and vector control using the novel tiny targets methodology. The idea was to create a model for integration of HAT control in the health zone. Most of the budget (3.5 million USD) was reserved for Memisa for support to the healthcare system, including renovation and construction of facilities. The Liverpool School of Tropical Medicine (LSTM) were involved as partner to introduce and pilot the tiny targets approach, ITM supported the PNLTHA in introducing active case finding with mini teams. For this purpose, apart from the two hospitals, HAT microscopy was also introduced in a number of health centers. Unfortunately the new grant caused a lot of friction with the PNLTHA management at the time, who were not satisfied with their role as subrecipient from ITM. They insisted that funds should go directly to them and that they would decide on which activities were to be implemented and how. The idea of integration of HAT control activities in the health zone, with the health zone management team in charge, appeared difficult to accept.

Screening activities in Beno were completed during the 4th quarter of 2014 but the confirmation visits suffered some delay. Early 2015 the lab technician visited 179 suspects, positive on either CATT or RDT, with or without positive result on LAMP. The lab technician was able to retrieve about 80% of these, among whom 53 CATT positives. Two were confirmed as HAT cases, both CATT and RDT positive but surprisingly LAMP negative. Two good quality videos were recorded, one of a LNA, one of a mAECT (Animation 1). The filter paper samples collected during initial screening and at the time of confirmation were tested with ITL at ITM. In both series there were two positives, on both occasions these were the filter papers belonging to the two cases. The positive predictive value of CATT was approximately 4%, very different from the 64% reported by traditional mobile teams in the same area.

Animation 1: Videos of confirmed HAT cases.

HAT Confirmed Case BENO 1.mp4   HAT confirmed case BENO 2.mp4

In Mbaye Larème the set up with 3 motorcycle based screeners followed by a motorcycle based confirmer after a delay of one week worked out well. Thus we decided that this would be the model to implement in Yasa Bonga and Mosango.

In spring 2015 results from a phase 3 evaluation of the RDT from Standard Diagnostics (SD) became available, in which the SD RDT had been compared to CATT. Surprisingly the sensitivity of CATT was far inferior to that of the RDT (69% vs. 92% and highly significant) [37] Because of earlier doubts about the quality of diagnostic confirmation we decided to conduct a similar study but with videos of positive confirmation tests as QA. We selected the two mobile teams that had identified the highest numbers of cases the year before, Kutu and Bolobo. Surprisingly in 5 months' time one of the teams did not find a single case, the other found 6 but was not able to record any useable videos because they forgot to put in the ocular tube. Also the 6 cases were found just when the study supervisor who shuttled between the two teams was absent.

In July, 2015 a new grant was signed (OPP1136014, 7.4 million USD, named 'TrypElim') with as objectives to fully cover the health zones of Yasa Bonga and Mosango with active screening by mini-teams, also extending coverage to villages considered non-endemic but situated within 5 kilometer of an endemic village, extending vector control with tiny targets across the entire area of the two health zones, implementing passive case finding for HAT in all health centers based on a system with RDT screening and referral to hospitals or health centers equipped for diagnostic confirmation. In addition the electronic data collection and QA system was to be further developed, for which E-Health Africa were involved as additional partner.

Another study was done by PNLTHA and FIND later that year, this time with an RDT based on recombinant antigen. Again the result was surprising to say the least, this time the recombinant RDT was most sensitive (71.2%), followed by CATT (62.5%) and by the original RDT, SD Bioline HAT with 59.0%. [38] These results further affirmed our suspicions of problems with the routine diagnostic confirmation tests in mobile teams. Such problems had already been experienced by MSF and gracefully acknowledged in a report published online. [10] In the meantime we had further refined the system by using a camera that can be inserted directly in the ocular tube of the microscope and connected to a smartphone or tablet. [39] Unfortunately it took a lot of time for this innovation to be embraced by the PNLTHA. The then director

(Crispin Lumbala) and his senior advisor (Pascal Lutumba) were not in favor, citing as reason that this would amount to a lack of trust in the skills of field staff. For others it was clear though that there was a problem and in January, 2016, DNDi organized a refresher training for all staff involved in microscopy in mobile teams.

In November, 2015, a dissemination meeting took place of the first grant. In the meantime talks had begun about further expansion of activities. Matt Steele introduced us to Trad Hatton of PATH, who was based in Kinshasa. Since the relation with the then director of the PNLTHA and his senior advisor had soured so much there was a need for someone able to lobby with their superiors to prevent them from further obstructing the process. In addition ITM at the time did not have an office or representative in Kinshasa and PATH was considered a strong partner in management and logistics.

In July, 2016, a next grant was signed (OPP1155293, 14.1 million USD, named 'TrypElim Bandundu'), with the main aim of extending the activities all over the former Bandundu province (now split in Kwilu, Kwango and Mai Ndombe), home to more than 50% of all HAT cases across the continent at that moment. The idea was to increase screening coverage in a systematic approach, through additional mini-teams moving from south to north and covering all endemic health areas over a 3-year period. The traditional mobile teams would continue with their regular approach. The electronic system was to be further developed, including a dashboard and planning tool. PATH were added as a partner for logistics and financial management, in particular to organize payments of field staff making use of 'mobile money'. LSTM were to expand vector control by tiny targets across suitable areas all over Bandundu province. The role foreseen for the PNLTHA was to prepare for expansion of the systems developed across the country.

Over the course of the first year of implementation of the TrypElim grant (OPP1136014) there were some issues with the development of the electronic data collection system and dashboard. Development and implementation were lagging behind and in consultation with Matt Steel it was decided to have E-Health Africa replaced. After a thorough selection process Bluesquare, a Brussels based SME, took over.

In April 2017,at the occasion of the 5th anniversary of the London declaration on NTDs, Bill Gates and the Belgian Minister Alexander De Croo in charge of development cooperation, jointly announced in Geneva that they would take the lead in the elimination of HAT. A Memorandum of Understanding was signed for this purpose. An additional Belgian grant was made available, called 'HAT plus', for an amount of 9 million Euro. To prepare this, a delegation of BTC and ITM led by Paul Verlé (at the time still with BTC), accompanied by Wim van Damme and Epco Hasker visited Kinshasa to meet with the potential stakeholders. There was a clear instruction from DGD that funds could not go directly to the PNLTHA.

In 2018, a 4.2 million USD supplement was provided by the BMGF to OPP1155293, with the aim of further expanding vector control. An overview of all the grants to date is shown in figure 4 below.

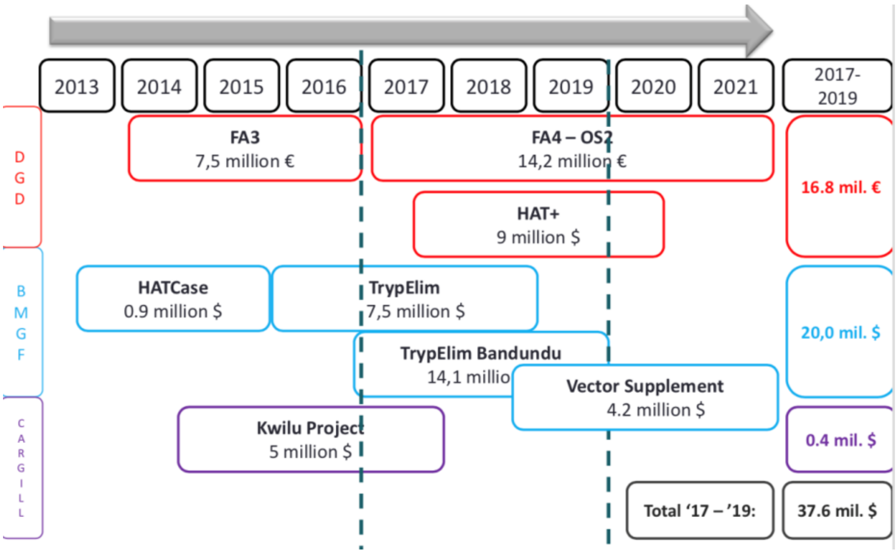

Figure 4: Overview of all grants to date

Relations with the PNLTHA director and his advisor continued to deteriorate. Implementation of the new grant (OPP1155293) was hampered by their refusal to share data and to allow involvement of ITM in the decisions on where to deploy the additional mini mobile teams outside Yasa Bonga and Mosango. As they were both related to the Minister of Health at the time it was very difficult to intervene. Only when a new minister was appointed in 2018 (Minister Ilunga), it became possible to move things forward. The new minister replaced the PNLTHA director and appointed the current director, Dr. Erick Miaka. Since then there has been considerable improvement but Dr. Miaka too has only limited maneuvering space. His position was insecure when in 2019 a new minister of health was appointed but for now he seems to have prevailed. He has signaled on several occasions that he is in favor of implementing changes required to adapt to the new epidemiological situation but has to maneuver very carefully. During a site visit in October, 2020, we observed that numbers of cases found had dropped significantly even though record numbers had been screened in 2019. Figure 5 below shows numbers screened (blue line) and cases detected (red bars) over the years.

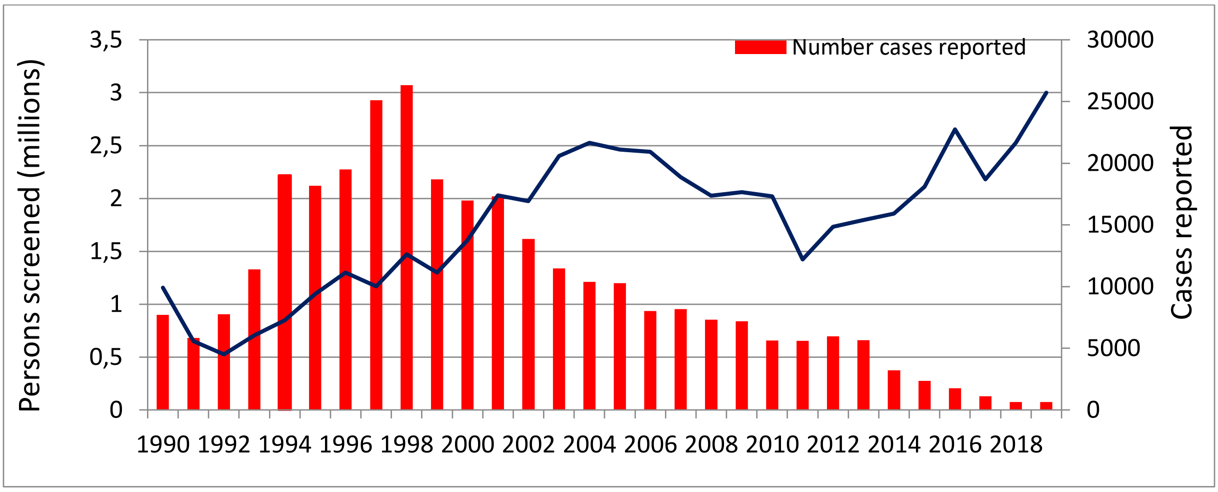

Figure 5: Numbers screened (blue line) and HAT cases detected (red bars) by year.

In 2019 approximately 3 million had been screened and 603 HAT cases were reported. However approximately half of those had been found through passive case detection. The yield of active case detection was thus approximately 300 on 3 million screened. Still there are 30 large mobile teams plus 18 mini teams, the average team diagnosing less than 10 cases per year. Reportedly some of the mini-teams were yet to diagnose their first case but contrary to what was the intention in the TrypElim Bandundu project, they had not been moved to other zones. Obviously rationalization of screening efforts is mandatory. It was agreed that ITM epidemiologist Anja De Weggheleire will come to Kinshasa in December, 2020, to compile a list of villages still in need of screening according to the WHO criteria (having reported cases in the past 3 years or 5 years ago) and make an estimate of numbers of large and mini-teams required to cover this population. On the positive side we observed that the electronic data collection and QA tools appeared to be accepted and working well, though there are still some outstanding issues related to data collected and uploaded in earlier years. The system is going to be extended to all mobile teams in the near future.

Recent developments
A few additional recent developments need to be mentioned. Apart from the new oral drugs, Fexinidazole and soon expected Acoziborole, new tests have also become available. Already in 2010 we had explored the possibility of using micro-CATT, an adaptation of the CATT test applied on dried blood spots (DBS) collected on filter paper.[40] We concluded that this was feasible but rather labor intensive. At the time we also tested ELISA/*T.b. gambiense* which also had good sensitivity and specificity. However with prevalences ever declining, positive predictive value was a major issue. Adding on Immune Trypalnolysis (ITL) as confirmation test could be a solution but ITL is not a field friendly test. A system based on ELISA/*T.b. gambiense* followed by ITL was piloted in Kongo Central recently and showed promising results. There were no ITL positives in over 5,000 samples from non-endemic areas and among samples from endemic areas only two out of 5,675 were ITL positive, one was a former HAT patient, the other died before further investigation had been done. Very recently a new inhibition ELISA (*g*-iELISA) has been developed at ITM with support from the BMGF. A phase 2 evaluation at ITM showed 92.6% sensitivity and 100% specificity on DBS samples. [41] Contrary to ITL, *g*-iELISA can in principle be performed in provincial level laboratories in DRC. The combination of ELISA/*T.b. gambiense* and *g*-iELISA, performed on the same DBS filter paper can be used for multiple purposes. It can be a tool to certify absence of transmission in formerly endemic areas, it can also be used as a screening tool in areas still endemic. Rather than bringing the laboratory expertise to the field, samples can be brought to a regional or central laboratory. If samples positive on both tests were to be found, this would still require sending a lab technician for confirmation and treatment. However with the tools developed over the years, finding back a village and a patient should not be a major issue. A diagnostic confirmation can then be performed and a blood sample can be taken for further testing by PCR at a remote laboratory. Once Acoziborole becomes available such persons can anyhow be treated, irrespective of whether or not HAT is confirmed.

Acoziborole will allow to move from a strategy based on screening, confirmation and treatment to simply screen and treat. However at present the positive predictive value of the screening routine tests used (CATT and RDTs) is in the order of 1-2% only.

Obviously for the combination of ELISA/*T.b. gambiense* and *g*-iELISA this would be much higher. Still there is a need to monitor on a continuous basis whether we are still treating truly HAT infected subjects or just false positives. For this purpose *g*-iELISA on top of CATT or RDT will be of great use since it only requires keeping a DBS sample on filter paper of each presumed case treated.

Another option is a newly developed PCR, currently in the final stages of validation. A venous blood sample collected in a special DNA/RNA shield tube can be stored at ambient temperature for several weeks before being processed in a remote laboratory. This is a highly sensitive and specific test that will provide evidence of whether or not DNA of *T.b. gambiense* is still circulating. A possible application of this test would be to ask mobile screening teams to collect a DNA/RNA shield tube from all confirmed HAT cases as well as one suspect a day in whom the presence of the parasite could not be demonstrated. This would allow monitoring not only the specificity of microscopy (already monitored by recording of videos) but also the sensitivity. In addition it would allow to establish a sample bank that can be highly relevant for future studies when it will become increasingly hard to still find HAT cases.

Also related to the RDTs there has been an important new development. Over the past couple of years there have been supply issues, partly related to a lack of interest from manufacturers to produce tests for which there is not a big market, partly related also to supply issues of the antigen required, which is produced at ITM in an artisanal process. An RDT based on recombinant antigen developed by Standard Diagnostics had been announced several years ago but is still not available.[38] Now recombinant versions of LiTat1.3 and LiTat1.5 have been developed at ITM and evaluated on a sample of 42 cases and 24 endemic controls. Areas under the curve of 1.0 for recombinant LiTat1.3 and 0.93 for recombinant LiTat1.5 were observed. If this can be validated on a larger sample it could once and for all settle the issue of antigen production for RDTs.

One more test that needs to be mentioned is an ELISA that detects antibodies against saliva of tsetse flies, also recently developed at ITM. The antibody response is short lived, so a positive test indicates recent exposure to tsetse flies. It is still in the process of further validation. Sensitivity and specificity are probably insufficient to reliably identify exposure at individual level but sufficient to identify group level exposure.

The proposed strategy for the coming years

With prevalence of HAT at such low levels and new tools that have become available now is the time to fundamentally change our approach to HAT control. To keep HAT in DRC under control and eventually interrupt transmission, the main pillars are:

- passive case finding
- active surveillance
    - active screening according to WHO strategy
    - reactive screening
    - probing of blind spots
    - surveillance of historic foci
- Treatment

- vector control

Passive case detection:
Passive case finding has over the years identified approximately 50% of all cases notified and this is still the case today. However experience in Mosango and Yasa Bonga has shown that there are some limitations. In these health zones RDTs had been made available in all health facilities and microscopy for diagnostic confirmation had been made available not only in the district hospitals but also in some health centers. But as it turned out, the vast majority of HAT suspects with a positive RDT identified in peripheral health centers did not present to a higher level facility for confirmation. The only places where HAT cases were diagnosed was in the hospitals. The few referred RDT positive suspects that did present for confirmation did so in the hospitals, even if a health center providing diagnostic confirmation testing was much nearer. (Snijders R.,2020, under review). With HAT prevalence further decreasing, screening with an RDT at health center level has become untenable. The positive predictive value has become so low that the vast majority of referrals are false positives. It makes more sense to concentrate passive screening and diagnostic confirmation in hospitals. Diagnosing HAT is not easy though and structures seeing less and less cases are bound to make mistakes. To avoid wastage of resources it is imperative that stringent QA procedures are put in place. QA can be either in the form of videos of positive tests or samples collected in DNA/RNA shield medium, to be confirmed by PCR, or filter paper DBS samples to be confirmed by *g*-iELISA. The presumed HAT patients can anyhow be treated but before reactive screening is implemented we need to be very certain about the diagnosis.

Once Acoziborole becomes available one could even consider treating on the basis of symptoms plus a positive RDT, without parasitological confirmation. However it is then imperative to have a mechanism in place to verify post-hoc whether these patients were not just false positives. The easiest option here is probably to collect a filter paper DBS sample to be tested with *g*-iELISA in a regional lab.

Active surveillance:
For active surveillance we foresee 4 different modalities: 1) continuation of the current WHO strategy requiring any village that reported a case to be screened until no more cases are found for three years in a row, with one additional round of screening five years after the last case was found, 2) reactive screening of villages from which cases were detected during passive screening, 3) probing of blind spots, i.e. active screening in areas with suspected transmission that have not been under regular surveillance, and 4) surveillance of historic foci, i.e. areas that have been endemic but have not reported any cases for at least five years.

1. Routine active screening: In continued mass screening efforts there is an urgent need to further rationalize the selection of villages to be planned for screening. This planning is made at the beginning of the year and in principle doesn’t change during the year. For this purpose we will use the SENSEHAT tool developed in the Trypelim project. There may still be some issues with completeness of data from previous years and the system is not yet in use all over the country but we must now make good use of it, learn the lessons to be learnt and fix any issues that may still become apparent. If the WHO criteria for eligibility of villages for screening are consequently applied (i.e. only select villages from which HAT cases have been reported in the last

three years or five years ago), there is no doubt that the total population to be screened will drop significantly, as early as next year. A decision will have to be taken on what to do with the 30 large mobile teams and 18 mini teams currently still active.

2. Reactive screening: There are also villages from which passively detected cases have been reported that are not planned for screening because they are considered non-endemic. Cases from such villages need to be confirmed and if so, reactive screening needs to be put in place. Such reactive screening needs to be planned on a continuous basis, throughout the year, whenever cases are reported from villages not already planned for screening later during the year. Reactive screening is probably much more effective if carried out by a mini-team that can also easily be transferred from one area to another. Currently mobile teams belong to a health zone and moving the equipment, the vehicle in particular, to another health zone is usually not allowed by the population. This issue can be overcome by having mini-teams at provincial level that can be deployed wherever there is a need. The current model of 3 motorcycle based screeners followed by 1 motorcycle based microscopist is working but can be further improved. As shown in figure 6 below, participation in door-to-door screening is better with mini teams than with large teams, particularly among adult men. To avoid delays between screening and confirmation, teams can be reduced to two persons of which one does screening only, the other screening as well as confirmation.

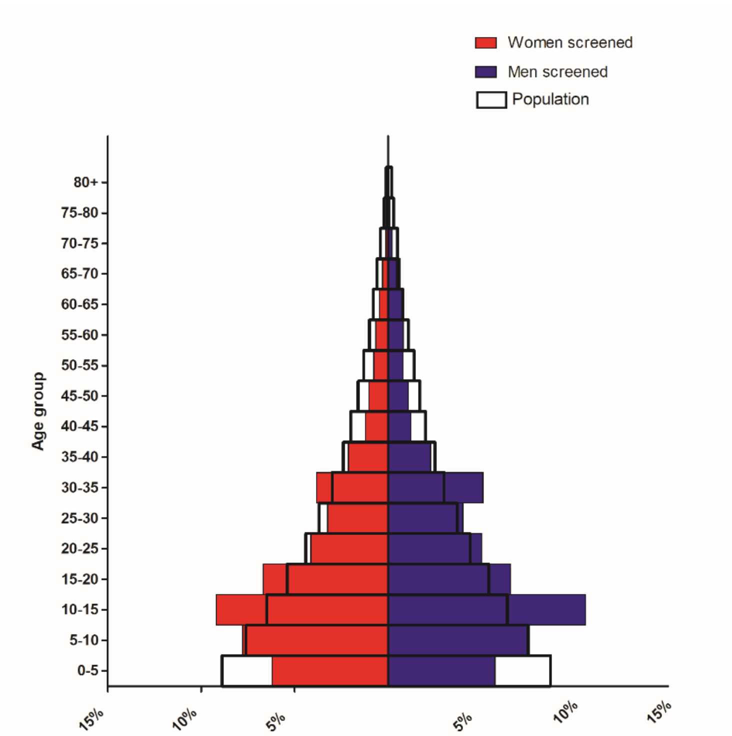

Figure 2: Participation in HAT screening by mini teams of women (red bars) and men (blue bars). The transparent bars show the population distribution.

3. Probing of blind spots: Areas that have not been under regular surveillance but are suspected to be potentially HAT endemic need to be screened at least once to exclude ongoing transmission. This can also be done by mini-teams used for reactive screening but another option would be to collect DBS samples on filter paper for

testing at a provincial level laboratory making use of the combination of ELISA/*T.b. gambiense* and *g*-iELISA. In case *g*-iELISA positives are found they would need to be revisited for diagnostic confirmation. A mini-team could be dispatched for this purpose that could also expand the probing further if cases are found.

4. Surveillance of historic foci can be based on a similar approach as used for probing of blind spots. They do not need to be screened every year but perhaps once in five years. It needs to be borne in mind that HAT is a focal disease, not finding any cases in one health area does not mean that the entire health zone is free of HAT. To exclude with 95% certainty a prevalence of 1 per 1,000 would require a sample size of 3,000, which is still manageable. If a resurgence can be detected at that stage it can still be controlled relatively fast. Aiming to detect prevalences in the order of magnitude of 1 per 10,000 is not feasible.

In support of active and passive surveillance there is an urgent need to expand mapping efforts beyond Bandundu, where all villages have been mapped in an effort coordinated by UCLA and financed by the BMGF. We do suggest however to adopt a much simpler approach. In the DHIS2 system lists are available per province and per health zone of all health areas (*aires de santé*) with population estimates. With these lists as a basis we can ask each (previously) HAT endemic health zone to provide the names of all villages per health area, typically 5-10, with estimated populations. The next step would be to verify for which of these villages we already have GPS points. Final step would be to provide health zones with a simple app on a smartphone that allows to select from a dropdown menu province, health zone, health area and village name and then press a button to record the GPS coordinates, while standing in the center of the village.

Treatment :
Effective use of fexinidazole is the challenge for the next years to come. Fexinidazole has the potential to simplify the diagnosis and treatment of gHAT. Following the recommendations of EMA, several limitative steps have been integrated into the new interim guidelines of WHO [41], which could lead to underutilization of its potential. However it is essential to collaborate with the ongoing pharmacovigilance system and to collect data of its use to contribute to the evidence and to the update of the guidelines in the near future. It is hoped that the availability of the new single-dose oral compound of acoziborole will also be added to the toolbox and also will require an update of the WHO guidelines.

Vector control:
The next element of HAT control is vector control based on the tiny targets method. In suspected sites of transmission, tsetse density needs to be determined, followed by deployment of tiny targets. Based on sound epidemiological data and modeling, further areas can be identified where tiny targets can be deployed. Xeno monitoring can be considered but should be piloted first since numbers of tsetse flies are generally very low and only a very small fraction carries *T.b. gambiense*. The yield and cost would need to be compared to that of surveillance of the human population.

Operational research
With less and less HAT cases found we need to constantly assess and adapt the approach used to keep the disease under control and eventually stop transmission.

Contact tracing is one of the elements to be considered. Even if the parasite requires a vector to be transmitted from one subject to another, exposure to infectious tsetse flies is probably shared between people. Collecting venous blood samples of confirmed HAT cases in DNA/RNA shield tubes will allow not only PCR for confirmation of parasite DNA but also DNA sequencing. In a pilot study about to be initiated we are now going to validate this approach and hope to soon find out how much diversity there is in parasite strains and whether clusters can be identified. In case a cluster of patients is identified we will search for other links and possible common places of exposure. For this purpose qualitative research is required to establish social networks to be related to potential places of tsetse exposure. In our pilot study we will also use GPS trackers to document potential transmission sites visited by recent HAT patients. The anti-tsetse saliva ELISA will also be used to identify exposure to tsetse flies at group level, e.g. inside a village or among people gathered at a river crossing.

The *g*-iELISA has performed very well in a phase 2 evaluation but before we can actually use it as a surveillance tool a further phase 3 evaluation is urgently required. Such a study could be combined with further validation of the PCR and setting up of a sample bank as explained below.

We also need to study potential alternative reservoirs of *T.b. gambiense* such as animals or healthy skin carriers. As human HAT cases become rare, such alternative reservoirs could become more important. This is a research question that needs to be answered before interruption of transmission can be certified. Similar issues are at play in elimination of visceral leishmaniasis from the Indian subcontinent and important synergies can be created.

With numbers of patients detected below 1,000 per year, now is the time to establish a sample bank for future research. It should not be impossible to collect in DNA/RNA shield tubes samples for all confirmed cases. Thus we could not only establish a sample bank for future research but it would also allow monitoring whether the numbers of circulating strains are diminishing.

To summarize, for the period until 2030 we foresee rationalization of (re)active screening, surveillance of absence of transmission in previously endemic areas and exploratory surveillance in areas of unknown endemicity as key elements. Strict quality assurance needs to be implemented, both in active and passive case finding. Vector control based on tiny targets can reinforce these measures in well targeted areas. As cases become fewer, now is the time to collect and preserve as many samples as possible and start experimenting with more advanced tools such as sequencing.

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